

Tranquilytics

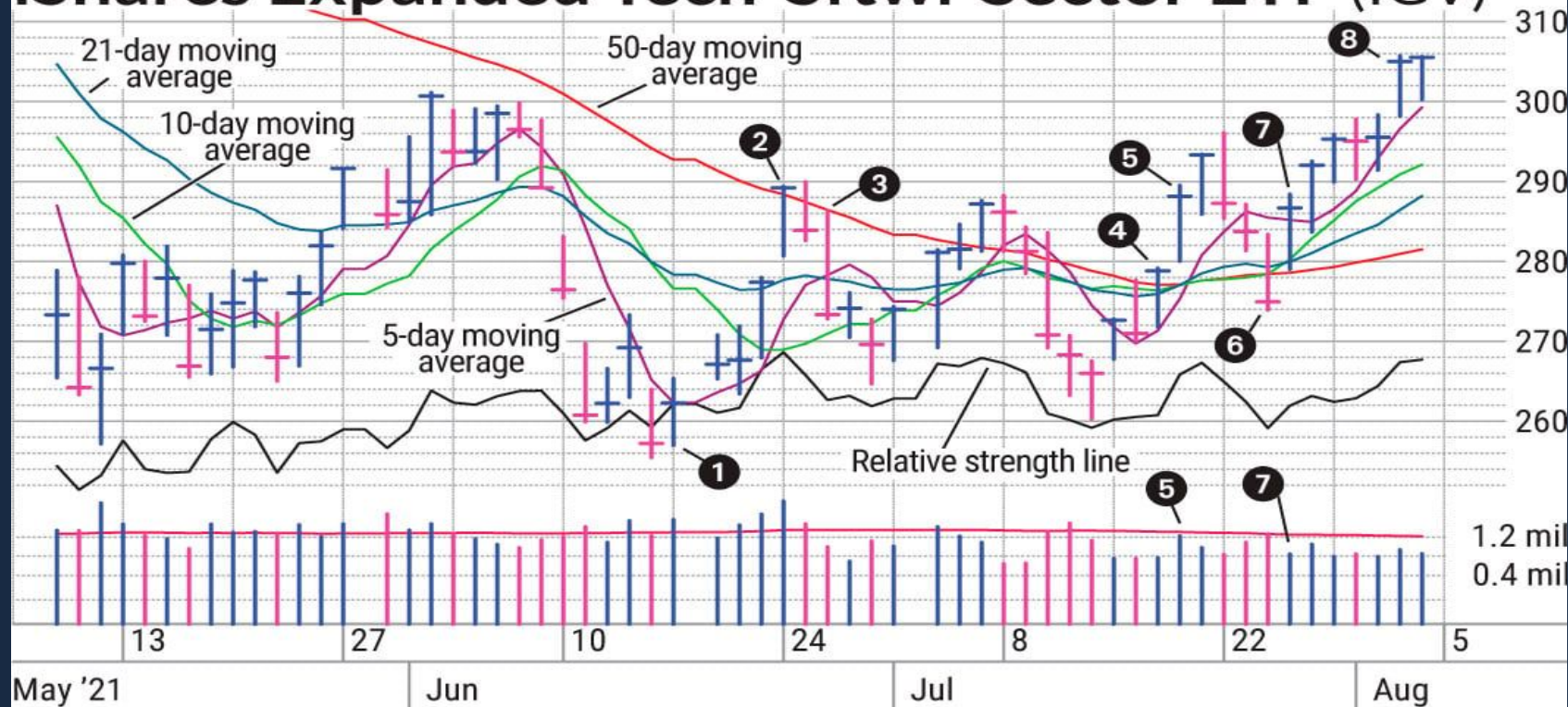
Stress Free AI Stock Analysis for
Anyone

By Johann Jaramillo
CSC 492 - Senior Project
Spring 2026

California State University Dominguez Hills



iShares Expanded Tech-Sftwr Sector ETF (IGV)



Why Tranquilytics?

Successful Stock Investing is
DIFFICULT!



1.

Starting out, Investing feels
like gambling

2.

Steep learning curve with
high risks requiring extensive
knowledge

3.

Research is a must, quality
information is not easy to
gather

4.

Current tools are usually:

- A. Expensive
- B. Complex
- C. Subpar

The Solution

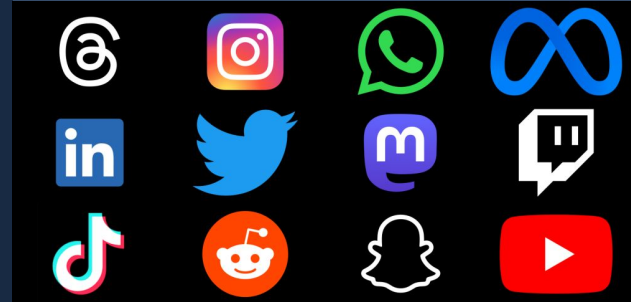
An AI-driven Stock Analysis Platform

Provides a low-cost, accessible platform to inform users on decisions using:

- Market Data (Historical analysis)
- Public Sentiment
- Statistical Analysis

Presented on a Dashboard that shares:

- Overall Stock Scores
- Clean Reports
- References



Utilizing Three Specialized Artificial Narrow Intelligence [ANI] Agents to Gather and Synthesize Information

Technical Agent

- Price History
- Volatility/Risk
- Statistical and LSTM Algorithmic Analysis
- Assigns a Technical Score



Sentiment Agent

- Uses Natural Language Processing (NLP) & Logistic Regression
- Data Scraped from Social Medias, Forums, and Newsletters
- Assigns a Sentiment Score



Synthesizer Agent

- Fed Data from Previous Two Agents
- Weighs and Balances the Findings using Backprop
- Assigns a Final overall score
- Generates a Stock Report for the User



Final Result

- Daily Reports Generated
- Balanced Analysis (Logical Data + Market Feel)
- Two Clicks Tell you Everything
- Simple, Concise Presentation of Information
- Third Click Provides References and Reasoning



System's 5-Step Process

1. Ticker Input

User Requests inputs ticker symbol

Search bar available on main dashboard page for quick lookups

2. Ticker Validation

External API Validates User Request

Proceeds to Step 3 if Found, suggests closest tickers if not.

3. Collect Data

Scrape Data from Relevant Websites

Gather and store information, ideally minimum 7 sources for logical and sentimental evaluation



4. AI Processing

AI given ticker data from DB

The three agents combined logic provides a conclusive summary (Longest Process)



5. Final Report

Display Interactive Summary Page

Final Summary Pops up, featuring each agent's ideas, the overall score (short+long term), expandable option to view sources

MSFT 5K@ 61.25 ▼ 1.35

↑
TICKER
SYMBOL

↑
SHARES
TRADED

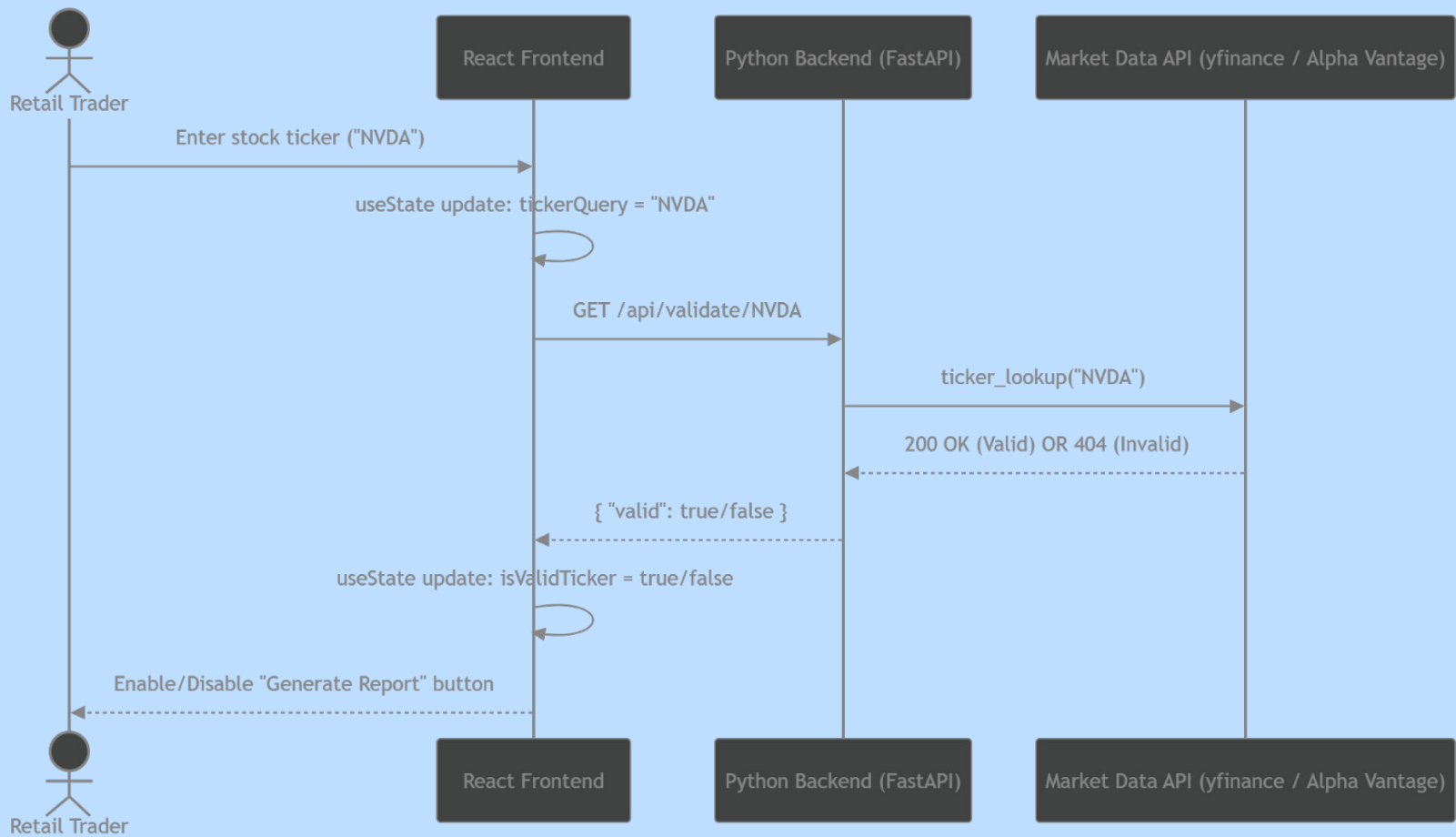
↑
PRICE
TRADED

↑
CHANGE
DIRECTION

↑
CHANGE
AMOUNT



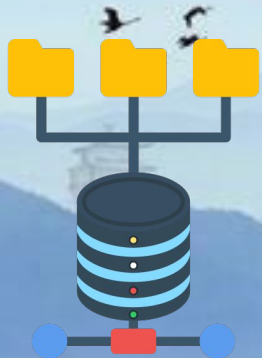
Alpaca



Sequence Diagram for Stock Ticker Lookup

Key Features

Multiple Sources Processed
through different
reasonings



Transparent, Clear Analysis
providing sources, weights,
and biases.



Intuitive, Fast results
designed for everyday
investors, *not* experts.



Free/Low-Cost approach,
sustainable for long-term
usage



Related & Future Work, Tech Stack

Related Work:

30% Profit Returns as a 2 year trader
CSC 411 - Artificial Intelligence
AI Tutor developed during CSC 481



Future Work:

Direct System Implementation, Controllers
and Custom APIs for AI

AI Training using Paper (Digital/Fake) Trading

Acquire Successful Prediction Accuracy of
>51%

Finalize Design for interactive User Dash
System Testing, Requirements Evaluation



Tech Stack:

Python - Flask, Pydantic, Pandas+Numpy

React+Tailwind CSS

Open-Source AI Model (Hugging Face) OR Gemini Flash
API

[Supabase.db](#) - PostgreSQL tables w/ 30 day cache

Vercel & Render for Hosting React and Py Web services

GitHub - Version Control

yFinanace, AlphaVantage - Stock Market lookup APIs



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